

Joel Teply

Principal ML Systems / Research Engineer

LLM Architecture · Inference · Model Efficiency · Continual Learning

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PROFILE

I design model architectures and the runtimes that serve them. Current work is transformer plasticity, structured pruning, and continual learning; the same specialty runs back through CNN quantization in 2017 and metric-learning architectures before AlexNet.

Twenty years of making research-grade computation practical on constrained hardware, which in this field means the model, the kernel, the memory system, and the scheduler are one problem rather than four.

CURRENT RESEARCH & SYSTEMS

Sentinel-AI — transformers that grow their own architecture · 2025–present

Architecture research with a written paper, *Experiential Plasticity*: attention heads that contribute to a domain specialize while the rest are removed, so architecture co-evolves with training instead of being fixed at design time.

- Qwen3.5-4B forged on code: **24% perplexity improvement over baseline from a smaller model**. Qwen3.5-27B: 3.5% better than baseline, running in 17 GB at 4-bit against 28 GB at fp16.
- LoRA with mixed-precision training; memory-tier detection so one recipe runs on a laptop or a 32 GB GPU. Models published on HuggingFace with the recipes that produced them.

github.com/CambrianTech/sentinel-ai

Model compaction — utilization-aware surgery, not blind quantization

- Head pruning by gate-gradient utilization: 14B coder model, 27 GB → 8.9 GB (3×).
- MoE expert pruning by runtime activation profiling: 35B, 67 GB → 47 GB.
- Both published with reproducible recipes; compacted models run on hardware that could never hold the original.

Continuum — distributed local-first AI · 2025–present

Open-source Rust platform for persistent AI agents across commodity hardware. Sole architect, 550,000 lines.

- **Inference runtime.** 121 commits on a maintained llama.cpp fork: backend lifetime contracts, slot and KV-cache behaviour, Metal and CUDA placement. ~50 tok/s solo and ~128 batched on Metal, 80–237 on consumer CUDA; a 48B linear-attention model at ~57 tok/s on a Mac.
- **Scheduling under memory scarcity.** Typed KV-slot leases, traffic classes and priced eviction so a dozen agents share a few warm slots instead of each holding a model.
- **Continual learning, evaluation-gated.** Agent experience generates LoRA training curricula; an adapter is adopted only when evaluation improves. 43 of 76 resolved SWE-bench Verified attempts (57%), with reproducible verdict artifacts.
- **Distributed training and inference.** Peer-to-peer mesh moving work and learned adapters toward suitable hardware, with signed provenance on produced artifacts.
- Fourteen concurrent embodied agents at 30 fps with real-time voice, rendered by the same machine serving their models.

github.com/CambrianTech/continuum

EXPERIENCE

Cambrian Tech — Co-founder, CTO, Principal Engineer · 2009–present

Co-founded a computer vision and augmented reality technology company building proprietary perception technology and delivering engineering programs for enterprise clients across automotive, retail and home improvement. Owned technical architecture end to end and set technical direction across engineering teams. Technology licensed to Fortune 500 customers including Home Depot and Lowe's, deployed to millions of users. Automotive computer-vision program with Garmin, 2020–2022, under strict accuracy and CPU/GPU constraints, expanding into a multi-year engagement.

Applied ML, 2009–2022

- **Metric learning before AlexNet.** A modified Siamese network learned a patch-similarity metric driving cellular-automaton region evolution for segmentation. On our task it outperformed ImageNet-pretrained semantic segmentation and ran far faster — adaptive to the image, evaluating a growing frontier rather than dense inference. Migrated to CNNs once they generalized without a seed and had hardware acceleration behind them.
- **Quantization and compression, 2017.** Weight quantization and model compression tooling for deployed CNNs — the same problem now being solved at LLM scale.
- **Architecture retargeting.** ICNet adapted for both semantic segmentation and surface-normal prediction, trained and exported for on-device inference. Soft segmentation and matting for boundary quality.
- **Cross-framework engineering.** Caffe→TensorFlow and Caffe→CoreML/MLKit conversion so one trained model shipped to every target.
- **Synthetic data.** Image synthesis pipelines, including Unreal Engine scenes, generating labeled ground truth where real data was unobtainable.
- **Generative.** Inpainting, seamless texture synthesis, photo-driven video generation.
- **Deployment.** CNN inference at 1–3 second latency on phones, asynchronously revising a scene representation maintained at frame rate — segmentation and surface normals both on-device, no cloud.

First-named inventor, **US10964097B2** — *Pattern Recognition Systems and Methods for Performing Segmentation on Surfaces and Objects*.

EyeVerify / ZOLOZ — Software Engineer, Computer Vision & Biometrics · 2012–2016

One of the first two engineers on eye-vein biometric authentication; acquired by Ant Financial (Alibaba) for over \$100M in 2016. Vision algorithms at camera frame rate on constrained hardware; presentation-attack and liveness detection. Co-inventor, **US9665784B2**; also named on **US8437513B1** and **US8675925B2**.

TripleBlind → Ideem — Software Engineer, Cybersecurity · 2023–2026

Cryptographic authentication technology and the SDK surface enterprise customers build against.

Earlier

Propaganda3 (2010–2012) · **VML** (2006–2010) · **AgileWise** (2005–2006, hardware and embedded) · **NovaStar**, **Federal Home Loan Bank of Topeka**, **AllofE Solutions** (1999–2006)

TECHNICAL

ML architecture — transformers and LLMs, attention-head plasticity, structured pruning, quantization, LoRA and mixed-precision training, distillation, metric learning and Siamese networks, CNN architectures for segmentation/depth/normals/reconstruction, GANs, diffusion, synthetic data

Inference & systems — llama.cpp internals, KV-cache and slot scheduling, memory-tier planning, GPU compute (Metal, CUDA, OpenGL/GLSL), SIMD, concurrency, distributed systems, Rust, C++, Python, Go, TypeScript

Evaluation — benchmark harness design, reproducible experiment artifacts, provenance and attestation, telemetry that supports the claim

EDUCATION & PATENTS

B.S. Computer Engineering — University of Kansas

- **US10964097B2** (2021, Cambrian Tech LLC). First-named inventor.
- **US9665784B2** (2017), **US8437513B1**, **US8675925B2**.
- \$100,000 LaunchKC grand prize (2017); top-five finalist, Atlanta Startup Battle 6.0 (2019, TechSquare Labs).